A Lehmann–Romano Proof of Admissibility for Closed Convex Acceptance Regions

1 Lehmann–Romano supported formulation

The closed-convex acceptance-region argument in Lehmann and Romano, Section 6.7, gives a sufficiency result, not an unrestricted if-and-only-if result. The conclusion has two parts. First, a nonrandomized test with closed convex acceptance region is d-admissible, once the ray condition in Theorem 6.7.1 is verified. Second, if the same test has size exactly α at a finite null point, then Corollary 6.7.1 gives α-admissibility.

Thus the result proved below is

$$A_0 \text{ closed and convex} \quad \Longrightarrow \quad d\text{-admissibility},$$

and, with exact size attained at a finite point of the null,

$$A_0 \text{ closed and convex} \quad \Longrightarrow \quad \alpha\text{-admissibility}.$$

The exact-size condition is needed for the α-admissibility conclusion, but not for the d-admissibility conclusion.

2 The theorem and proof

Theorem 1 (Closed convex acceptance regions imply admissibility)**.** *Let* $0 < \alpha < 1$*. Let*

$$X \sim N(\mu, \Sigma),$$

where Σ *is known and positive definite and* $\mu \in \mathbf{R}^k$*. Consider testing*

$$H_0 : \mu \in \Lambda_0 = \{\mu : \mu \leq 0\} \quad \textit{versus} \quad H_1 : \mu \in \Lambda_1 = \mathbf{R}^k \setminus \Lambda_0.$$

Let ϕ_0 *be a nonrandomized test with acceptance region*

$$A_0 = \{x : \phi_0(x) = 0\}.$$

Assume:

(i) A_0 *is closed and convex;*

(ii) ϕ_0 *has level* α*:*

$$\sup_{\mu \in \Lambda_0} E_\mu[\phi_0(X)] \leq \alpha.$$

Then ϕ_0 is d-admissible for testing H_0 against H_1. If, in addition, there exists a finite point $\mu_0 \in \Lambda_0$ such that

$$E_{\mu_0}[\phi_0(X)] = \alpha,$$

then ϕ_0 is α-admissible.

Proof. We use Theorem 6.7.1 and Corollary 6.7.1 of Lehmann and Romano. The relevant result is the following.

Proposition 1 (Lehmann–Romano)**.** *Consider an exponential family with density*

$$p_\theta(t) = C(\theta) \exp\{\theta' t\}$$

with respect to a σ-finite measure ν, with natural parameter space Ω. Let Ω_H and Ω' be disjoint nonempty subsets of Ω. Let ϕ_0 be a nonrandomized test of $H : \theta \in \Omega_H$, based on T, with acceptance region $A_0 \subseteq \mathbf{R}^s$. Suppose A_0 is closed and convex and that the following ray condition holds: whenever

$$A_0 \cap \{t : a't > c\} = \varnothing,$$

there exist $\theta^* \in \Omega$ and $\lambda_n \to \infty$ such that

$$\theta^* + \lambda_n a \in \Omega'.$$

Then any competing acceptance region A satisfying

$$P_\theta(T \in A) \le P_\theta(T \in A_0) \qquad \text{for all } \theta \in \Omega'$$

is contained in A_0, except for a ν-null set. Consequently, the test with acceptance region A_0 is d-admissible. If the test has size α and there exists a finite point $\theta_0 \in \overline{\Omega_H}$ such that

$$E_{\theta_0}[\phi_0(X)] = \alpha,$$

then the test is also α-admissible. The same conclusions allow randomized competing tests.

We now verify the assumptions of this result. Write the normal density in canonical exponential-family form. Put

$$\theta = \Sigma^{-1}\mu, \qquad \mu = \Sigma\theta.$$

With respect to the measure

$$d\nu(x) = (2\pi)^{-k/2} |\Sigma|^{-1/2} \exp\left\{ -\frac{1}{2} x' \Sigma^{-1} x \right\} dx,$$

the density of $X \sim N(\Sigma\theta, \Sigma)$ is

$$p_\theta(x) = \exp\left\{ \theta' x - \frac{1}{2} \theta' \Sigma \theta \right\}.$$

Thus $T(x) = x$, $\Omega = \mathbf{R}^k$, and

$$\Omega_H = \{\theta : \Sigma\theta \le 0\}, \qquad \Omega_K = \{\theta : \Sigma\theta \not\le 0\}.$$

We apply the Lehmann–Romano result with $\Omega' = \Omega_K$.

It remains to verify the ray condition. Let $a \neq 0$ and $c \in \mathbf{R}$ satisfy

$$A_0 \cap \{x : a'x > c\} = \varnothing.$$

Equivalently, $A_0 \subseteq \{x : a'x \leq c\}$. We claim that

$$\Sigma a \not\leq 0.$$

Suppose instead that $\Sigma a \leq 0$. For $\lambda > 0$, define

$$\theta_\lambda = \lambda a, \qquad \mu_\lambda = \Sigma\theta_\lambda = \lambda\Sigma a.$$

Then $\mu_\lambda \leq 0$, so $\theta_\lambda \in \Omega_H$. Under P_{θ_λ},

$$a'X \sim N\left(\lambda a'\Sigma a,\ a'\Sigma a\right).$$

Since Σ is positive definite and $a \neq 0$, $a'\Sigma a > 0$. Hence

$$P_{\theta_\lambda}(a'X > c) \to 1.$$

Because $A_0 \subseteq \{x : a'x \leq c\}$, we have $\{x : a'x > c\} \subseteq A_0^c$. Since ϕ_0 is nonrandomized,

$$E_{\theta_\lambda}[\phi_0(X)] \geq P_{\theta_\lambda}(a'X > c) \to 1.$$

This contradicts the level condition because every $\theta_\lambda \in \Omega_H$ and $\alpha < 1$. Therefore $\Sigma a \not\leq 0$.

It follows that $\lambda\Sigma a \not\leq 0$ for every $\lambda > 0$. Hence

$$\lambda a \in \Omega_K \qquad \text{for every } \lambda > 0.$$

The ray condition holds with $\theta^* = 0$ and any sequence $\lambda_n \to \infty$.

The Lehmann–Romano proposition now implies d-admissibility of ϕ_0.

Finally suppose there exists finite $\mu_0 \in \Lambda_0$ such that $E_{\mu_0}[\phi_0(X)] = \alpha$. Let $\theta_0 = \Sigma^{-1}\mu_0$. Then $\theta_0 \in \Omega_H \subseteq \overline{\Omega_H}$ and

$$E_{\theta_0}[\phi_0(X)] = \alpha.$$

The second part of the Lehmann–Romano corollary therefore gives α-admissibility. □

3 Why exact size is needed for α-admissibility

The statement "closed and convex acceptance region implies α-admissibility" is false if one only assumes

$$\sup_{\mu \leq 0} E_\mu[\phi(X)] \leq \alpha.$$

For example, the zero test $\phi(x) \equiv 0$ has acceptance region $\mathbf{R}^k$, which is closed and convex, and it satisfies the size constraint for every $0 < \alpha < 1$. But it is not α-admissible. The test

$$\psi(x) = 1\{x_1 > \sqrt{\Sigma_{11}}\Phi^{-1}(1-\alpha)\}$$

has level α for $H_0 : \mu \leq 0$ and has strictly positive power at alternatives with $\mu_1 > 0$, while the zero test has zero power everywhere.

This example does not contradict d-admissibility. A d-dominating test would also have to satisfy

$$E_\mu[\psi(X)] \leq E_\mu[\phi(X)] = 0 \qquad \text{for all } \mu \in \Lambda_0,$$

which forces no increase in rejection probability on the null. Thus exact size is needed for the α-admissibility conclusion, not for the d-admissibility conclusion.

References

[1] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Third edition. Springer.